Alejandro Ortiz Perdomo

MLOps Engineer | Machine Learning Engineer

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PROFESSIONAL SUMMARY

MLOps Engineer focused on taking ML models and LLMs **from data to production**: data pipelines, reproducible training, deployment and monitoring. Implemented **CI/CD** with **Databricks + MLflow** that cut deployment from 2 weeks to **2 days**, and **data/model drift** monitoring with automated retraining. Experience deploying services on **Azure, AWS and GCP** with **Docker**, **Kubernetes (EKS)**, **Terraform** and serverless (**Azure Functions**, **Lambda**, **Cloud Run**), automated with **GitHub Actions**.

WORK EXPERIENCE

AI Engineer - AI Practice

Periferia IT Group
06/2025 - Present

- **Production CV-Job semantic search pipeline:** Took to production the ingestion (**Azure Document Intelligence**) and vector indexing (**Azure AI Search**) pipeline, packaged as **Dockerized Azure Functions** with SQL-triggered incremental indexing and scheduled sync. Processes **500+ CVs/day** and serves **10k+ searches/month** at **<200ms** latency.
- **Multi-cloud LLM agent platform (SRPA):** Technical design and documentation of a *plug-and-play* agent platform automating the SDLC, with LLMs such as **DeepSeek R-1** served on **Azure AI Foundry** and run on **Azure / AWS / GCP** for internal and commercial use; includes automated unit-test generation with >=**85%** coverage.
- **Production reference architectures:** Define architectures, estimates and team sizing to take **GenAI, Agentic AI and RAG** solutions to production for enterprise clients (banking, insurance, retail, energy and telco).

Freelance AI Engineer - Dental Clinic

Freelance
07/2026 - 08/2026

- **Serverless LLM agent deployment:** Took two agents (**LangGraph** + **Azure OpenAI**) to production on **Azure Functions** with queue-based processing (poison-queue handling and deduplication), Table Storage persistence and **Key Vault** secrets via managed identity.
- **Integrations & observability: WhatsApp Cloud API** webhook with HMAC verification, **Outlook (Microsoft Graph)** calendar with retries and token rotation, and **Application Insights** alerting. Deterministic critical logic (locking against double booking) and test-driven development with **118 tests**.

Machine Learning Engineer

Global Hitss
09/2023 - 06/2025

- **End-to-end MLOps on Databricks:** Took models from data to production: **PySpark** processing on **Delta Lake** governed by **Unity Catalog**, orchestration with **Databricks Workflows**, versioning and promotion in the **MLflow Model Registry**, and **endpoint** or **batch** serving depending on the use case.
- **Model CI/CD & operations:** Implemented the CI/CD pipeline that cut deployment from 2 weeks to **2 days**, with automated drift monitoring and retraining.
- **Production RAG agent:** Deployed a RAG conversational agent (**LangChain**, **Azure OpenAI**, **FAISS**) serving **10k+ queries/month**.
- **Models in production:** Productized the *Next Best Offer* system (Reinforcement Learning) with rollout validated by **A/B testing**, and the churn model with scores fed into the **CRM** for **50k+ customers**.

Machine Learning Consultant
Grupo CINTE
08/2023 - 09/2023

- **ML Observability:** Designed the monitoring methodology for *data/model drift* detection with automated retraining alerts, adopted as the data science team's standard.
- **Inference optimization:** Audited and optimized LightGBM and Random Forest models, reducing inference times and compute cost.

Data Scientist
NAE
03/2023 - 07/2023

- **Contactability model in operation:** Built and put into operation a **LightGBM** model prioritizing time windows and channels for campaigns of **100k+ contacts/month**.

Data Scientist II
Xolit
05/2022 - 01/2023

- **Batch production deployment:** Took models to production as **batch** jobs orchestrated with **Apache Airflow** on a defined schedule, delivering scores to the collections operation.
- **Production model monitoring:** Built **Power BI** dashboards tracking production model performance metrics and collection KPIs.
- **Models in operation:** Payment propensity (feeding **automated management priorities** for the portfolio), optimal contact time-slot recommendation and clustering-based account-holder segmentation.

EDUCATION

Master in Software Engineering
Universidad de los Andes
2025 - In Progress

B.S. in Statistics
Universidad del Valle
2011 - 2016

- Data Science Diploma (DS4A) - Correlation One, 2022
- Deep Learning Diploma - Univ. Autónoma Occidente, 2023

CERTIFICATIONS

- AI-102: Microsoft Certified Azure AI Engineer Associate - Microsoft, Feb 2026
- NLP Specialization (RAG/Transformers) - Coursera, Jun 2025
- PyTorch Ultimate - Coursera, Mar 2025
- MLOps and Databricks - Netec, Dec 2024
- AWS Cloud Practitioner - Udemy, Feb 2022

TECHNICAL SKILLS

- **GenAI & LLMs:** LangChain (Agents, RAG), LangGraph, Azure OpenAI (GPT-4, Embeddings), Prompt Engineering, Vector DBs (FAISS, Qdrant, Pinecone), LLM Evaluation (RAGAS)
- **Azure AI & Cloud:** Azure AI Search, Document Intelligence, Azure AI Foundry, Azure Functions, Databricks + MLflow (Workflows, Model Registry), Delta Lake, Unity Catalog
- **Google Cloud Platform (GCP):** Vertex AI (Training, Endpoints, Pipelines), Cloud Run, Pub/Sub, Compute Engine (MIG zonales), Cloud Storage, Cloud SQL
- **Amazon Web Services (AWS):** EKS, Lambda, DynamoDB, SNS, SQS, RDS
- **ML/DL Frameworks:** PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, Hugging Face Transformers
- **MLOps & Engineering:** Python, SQL, PySpark, Docker, CI/CD, Apache Airflow, Kubernetes, Terraform, Go, Angular, Kotlin, FastAPI, Streamlit, Git, GitHub Actions

SOFT SKILLS

Effective Communication, Problem Solving, Analytical Thinking, Team Collaboration, Adaptability, Continuous Learning, Time Management, Technical Leadership

LANGUAGES

- Spanish: Native
- English: Professional Proficiency (B2)

PROJECTS

RAG Assistant with Web Scraping and RAGAS Evaluation

github.com/alejortizp/rag-banco

Conversational assistant over a bank's public website: robots.txt-compliant crawler, **Qdrant** indexing, cross-encoder reranking and per-session memory. Evaluated with **RAGAS** (50 questions, independent LLM judge): faithfulness **0.87**; the reranker improved context utilization by **+8 pts** (A/B).

Technologies: Qdrant, Sentence-Transformers, Cross-Encoder, Groq Llama 3.1, RAGAS, Streamlit, Docker

Email Agent with LangChain

github.com/alejortizp/email-agent

Intelligent agent that queries emails via IMAP, classifies with LLM, and generates contextual automatic responses. Reduces response time by **80%**.

Technologies: LangChain, Azure OpenAI, IMAP, Python, Prompt Engineering

NL-to-SQL/Pandas Agent

github.com/alejortizp/nl-to-code

Agent that translates natural language to SQL queries and Pandas operations. Includes code validation, secure sandbox execution, and automatic results visualization.

Technologies: LangChain, SQL, Pandas, Code Interpreter, ReAct Pattern

ML Pipeline with MLflow: 7-Model Benchmark

github.com/alejortizp/MLOps-Pipeline-

Reproducible multivariate regression pipeline comparing **7 models** (Ridge, ElasticNet, Random Forest, Gradient Boosting, SVR, XGBoost, LightGBM) with hyperparameter search (RandomizedSearchCV, 5-fold CV) and **MLflow** tracking. The best model (Gradient Boosting, R^2 **0.84**) cut MAE by **26%** versus the baseline and is serialized as an inference-ready sklearn pipeline.

Technologies: Scikit-learn, XGBoost, LightGBM, MLflow, Pandas, Joblib

MLOps Text Classification System

github.com/alejortizp/mlops-text-classification

MLOps system classifying customer-support messages (public Twitter dataset): **training**, **batch prediction** and **monitoring** pipelines with accuracy-threshold alerts, plus **champion/challenger** retraining that only promotes a new model if it beats the current one in the **MLflow Model Registry**. TF-IDF + CatBoost model with **0.85** accuracy.

Technologies: MLflow, Model Registry, CatBoost, Scikit-learn, NLTK, TF-IDF, Pandas

ACADEMIC PROJECTS

Teaching-Assistant Platform on GCP (Go + Angular)

MISW4204 Cloud Solutions Development - 2026 - Universidad de los Andes

Top contributor (**60%** of commits, team of 4). Migrated the system from Docker Compose to an **autoscaling MIG + Load Balancer** and then to serverless **Cloud Run + Pub/Sub**, with asynchronous AI-generated PDF reports. JMeter load tests: **158 rps** (p95 292 ms) with **0%** infrastructure errors; **84%** (Go) and **93%** (Angular) coverage; SonarQube A rating.

Technologies: Go (Gin), Angular, PostgreSQL, Cloud Run, Pub/Sub, Cloud SQL, OpenAI, JMeter, SonarQube, GitHub Actions

Event-Driven Delivery Marketplace on AWS

MISW4301 Cloud-Native App Development - 2026 - Universidad de los Andes
Led the AWS architecture and IaC for **11 FastAPI microservices** (hexagonal architecture) on **EKS**, with a serverless extension: **SNS -> SQS** with DLQs, **4 container Lambdas** and DynamoDB, all in **Terraform** deployed via GitHub Actions to 5 environments. Fixed cold starts and IAM permissions, reaching **76/76** E2E assertions with zero failures.
Technologies: AWS (EKS, Lambda, SNS, SQS, DynamoDB, RDS), Terraform, Kubernetes, FastAPI, Docker

Microservices Architecture Experiments

MISW4202 Agile Software Architectures - 2026 - Universidad de los Andes
Hypothesis-driven experiments for a booking system: asynchronous payment-anomaly detection in **538 ms** mean vs. a 2 s target (t-test, $p<0.001$) and **100%** Man-in-the-Middle attack detection with zero false positives (RS256 JWT with revocation, RSA-OAEP, integrity hashing, TLS).
Technologies: Flask, Redis Streams, Celery, Nginx, JWT, RSA, SciPy

QR Alarms: from UX Research to Android + Web App

UX Prototyping - MISO - 2026 - Universidad de los Andes
github.com/a-ortizp/alarmas-qr-app
Full product cycle: user research, **3 usability tests** (20 findings), Figma prototype and Material 3 design system (48 components); then a **22-screen** Android app (ML Kit QR scanning, alarms firing with the app closed) and Angular web app, with signed APK releases via GitHub Actions.
Technologies: Figma, Kotlin, Jetpack Compose, CameraX, ML Kit, Angular, GitHub Actions